

Proposed activity about the simulation of a rigid top

Luciano Muñoz, Juan David Salcedo Hernández

1 Symmetric top

Consider a symmetric top, i.e., a rigid body which has its lower end fixed at a point (say, the tip of a needle) and that has two equal moments of inertia, i.e., $I_1 = I_2$. In this case, the Lagrangian for the rigid body simplifies to

$$L = \frac{I_1}{2}((\omega^1)^2 + (\omega^2)^2) + \frac{I_3}{2}(\omega^3)^2 - mgl \cos \theta.$$

Rewriting the angular velocity in terms of Euler, one obtains:

$$L = \frac{I_1}{2}(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{I_3}{2}(\omega^3)^2 - mgl \cos \theta,$$

so that the Euler–Lagrange equations take the form:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = \frac{\partial L}{\partial \phi} \implies \frac{d}{dt} (I_1(\dot{\phi} \sin^2 \theta) + I_3 \omega_3 \cos \theta) = 0 \implies (I_1 \sin^2 \theta + I_3 \cos^2 \theta) \dot{\phi} + I_3 \dot{\psi} \cos \theta = I_1 b \quad (1)$$

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\psi}} \right) = \frac{\partial L}{\partial \psi} \implies \frac{d}{dt} (I_3 \omega_3) = 0 \implies I_3 \omega_3 = I_3(\dot{\phi} \cos \theta + \dot{\psi}) = I_1 a \quad (2)$$

Now, by conservation of energy, there is a constant $\tilde{E}$, so that

$$\tilde{E} = \frac{I_1}{2}(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + mgl \cos \theta, \quad (3)$$

with $\tilde{E} = \text{Energy} - \frac{I_3}{2}(\omega^3)^2$. Notice that Equation 1 together with Equation 2 imply that

$$\dot{\phi} = \frac{b - a \cos \theta}{\sin^2 \theta}, \quad (4)$$

meaning the motion of ϕ is actually completely determined by that of θ . In turn, substituting Equation 4 into Equation 2 yields

$$\dot{\psi} = \frac{I_1 a}{I_3} - \cos \theta \frac{b - a \cos \theta}{\sin^2 \theta}, \quad (5)$$

and using the same expression in Equation 3 gives

$$\dot{\theta}^2 \sin^2 \theta = \sin^2 \theta \left(\frac{2\tilde{E}}{I_1} - \frac{2mgl}{I_1} \cos \theta \right) - (b - a \cos \theta)^2. \quad (6)$$

Let $u := \cos \theta$. The reader may verify that the last equation then becomes

$$\dot{u}^2 = f(u) := (1 - u^2)(\alpha - \beta u) - (b - au)^2, \quad \text{where } \alpha := \frac{2\tilde{E}}{I_1}, \beta := \frac{2mgl}{I_1}.$$

At this point, we invite the reader to follow the arguments presented in [Spi10, From page 246], where it is shown that the top will have *looped* nutation between certain angles whenever the polynomial $f(u)$ has two roots inside the open interval $(-1, 1)$ and the ratio a/b lies *outside* the open interval defined by those two roots. If a/b is in the interval between these two roots, then one has a simple nutation. As a cautionary note, Spivak makes a mistake and says looped nutation occurs when the ratio a/b is inside the open interval defined the two roots, this is not true.

One can explore the characteristics of the roots of the polynomial $f(u)$ with GeoGebra to find adequate parameters α, β, a, b that allow for simple or looped nutation. Indeed, assuming we have fixed physical constants $I_1, I_3, mgl, \tilde{E}, a$ and b , we can solve for the initial velocities, provided we can find some θ_0 , as:

$$\dot{\phi}_0 = \frac{b - a \cos \theta_0}{\sin^2 \theta_0}, \quad (7)$$

$$\dot{\psi}_0 = \frac{I_1 a}{I_3} - \dot{\phi}_0 \cos \theta_0, \quad (8)$$

$$\dot{\theta}_0^2 = (\alpha - \beta \cos \theta_0) - \dot{\phi}_0^2. \quad (9)$$

Observe that the last equality says the initial condition θ_0 is actually constrained by the equation

$$\dot{\theta}_0^2 = \alpha - \beta u - \left(\frac{b - au}{1 - u^2} \right)^2.$$

Thus one requires that the right-hand side be *positive*, which restricts the values of θ_0 to those angles with $\cos \theta_0$ being mapped to a positive number by the function on the right-hand side. This qualitative analysis may be carried out with the help of plotting software such as GeoGebra (see Figure 1).

In order for the simulation to be computationally stable, one needs to use the conjugate momenta as initial conditions for the system. Since our analysis thus far makes reference to velocities alone, one needs to compute conjugate momenta using the mass matrix described in the theory. We can do this using a simple python script:

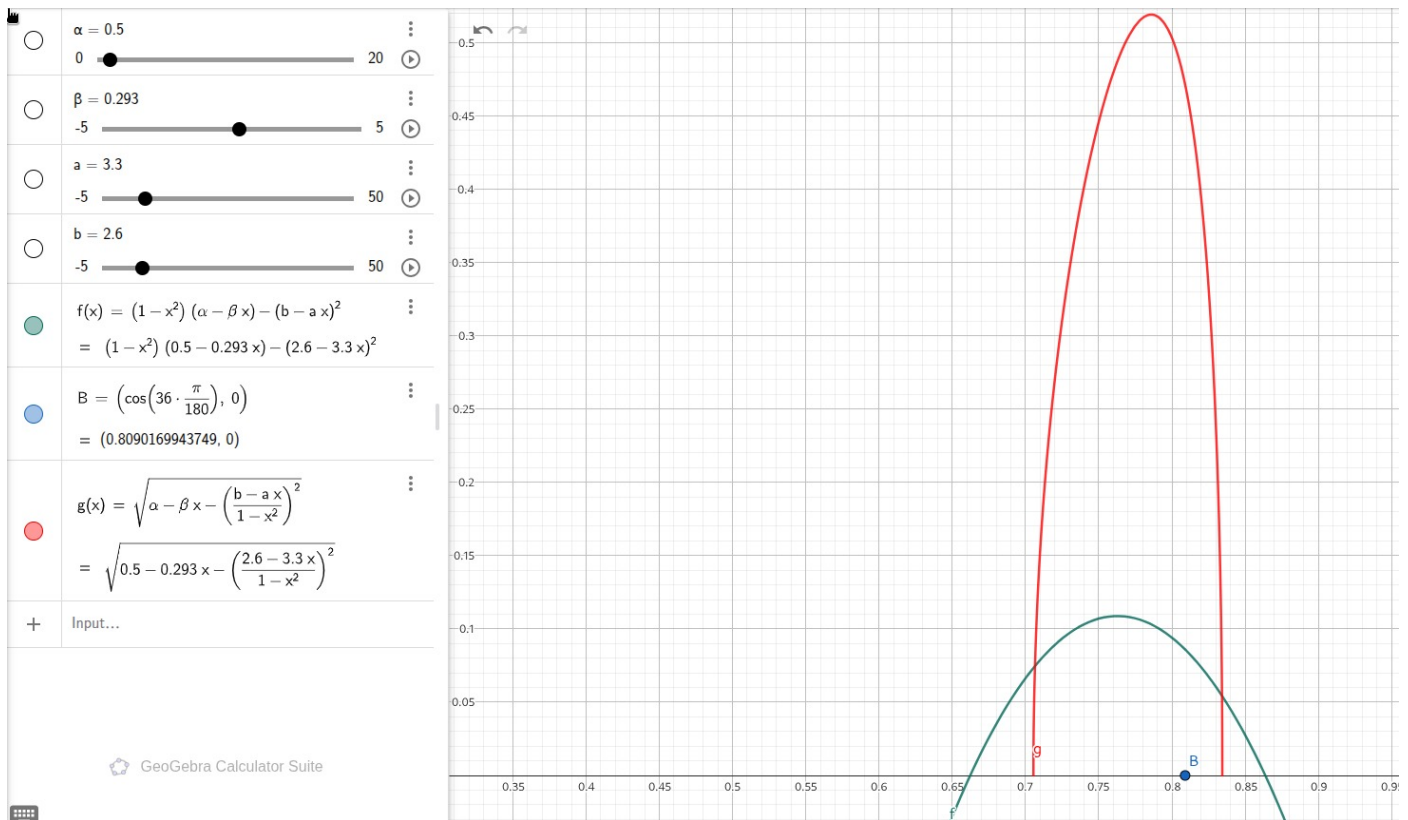


Figure 1. Process for qualitative analysis of top nutation.

```
import numpy as np

RAD = 180.0 / np.pi

def mass_matrix(th, ph, ps, I1, I2, I3):
    sTh, cTh = np.sin(th), np.cos(th)
    sPs, cPs = np.sin(ps), np.cos(ps)

    M11 = I1 * cPs**2 + I2 * sPs**2
    M12 = (I1 - I2) * sTh * sPs * cPs
    M13 = 0.0

    M21 = M12
    M22 = sTh**2 * (I1 * sPs**2 + I2 * cPs**2) + I3 * cTh**2
    M23 = I3 * cTh

    M31 = M13
    M32 = M23
    M33 = I3

    return np.array([[M11, M12, M13],
                     [M21, M22, M23],
                     [M31, M32, M33]])

def conjugate_momenta(theta0, a, b, I1, I2, I3, alpha, beta, phi=0.0, psi=0.0):
    # velocities from your equations
    phi_dot = (b - a * np.cos(theta0)) / (np.sin(theta0)**2)
    psi_dot = (I1 * a) / I3 - phi_dot * np.cos(theta0)
    theta_dot_sq = (alpha - beta * np.cos(theta0)) - phi_dot**2
    if theta_dot_sq < 0:
        raise ValueError("Negative value computed → no real solution")
    theta_dot = np.sqrt(theta_dot_sq)

    qdot = np.array([theta_dot, phi_dot, psi_dot])

    # build mass matrix
    M = mass_matrix(theta0, phi, psi, I1, I2, I3)
```

```

    # conjugate momenta
    p = M @ qdot
    return p*RAD

DEG = np.pi / 180

# Usage:
theta0 = 36 * DEG
alpha = 0.5
beta = 0.293
a = 3.3
b = 2.6
I1 = 2
I2 = 2
I3 = 1

p = conjugate_momenta(theta0, a, b, I1, I2, I3, alpha, beta)

print("Momenta p = [pθ, pφ, pψ] =", p)

```

For example, if one uses the constants $I_1 = 2 \text{ kg m}^2$, $I_3 = 1 \text{ kg m}^2$, $m = 0.1 \text{ kg}$, $l = 0.3 \text{ m}$, $g = 9.8 \text{ m s}^{-2}$, then we obtain $\beta = 0.293 \text{ s}^{-2}$; whereby, fixing $a = 3.3$, $b = 2.6$, $\alpha = 0.5 \text{ s}^{-2}$, one obtains precession. We invite the reader to verify this in the simulation. Also, the reader is prompted to explore the methods above to find other initial conditions. For instance, find conditions for simple nutation!